

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

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Micro- & Macroprudential Management · Country implementation of a CCyB increase

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Title (Zheng, 0:15) [0:00–0:15]

Good morning. We are Zheng Huang and Diego Gutiérrez, and our case is the Netherlands: how and why the Dutch central bank raised its countercyclical capital buffer from zero to two percent.

The Netherlands filled its CCyB to 2% when the rulebook said 0% — refilling a buffer, not fighting a boom

Decision: CCyB 0% → 1% (ann. May 2022) → **2%** (ann. May 2023, binding 31 May 2024), together with lower O-SII buffers.

Puzzle: the Basel credit-to-GDP gap stood at **-33pp** and **-47pp** at the two decisions → the buffer guide implied **0%**.

Q1 · Environment

Recovered economy, then war, energy shock and **11.6%** inflation; ECB hikes from Jul-2022.

Credit cold; housing hot (+19% y/y), then falling.

Banks strong (CET1 17.7%) and more profitable as rates rose.

Q2 · Criteria & risks

DNB framework (2022): **2%** = “**standard risk environment**”, calibrated on past crisis losses (€12bn), not on credit growth.

Insures against shocks no indicator predicts; housing left to other tools.

Keeps the March-2020 promise to restore releasable capital.

Q3 · Implications

Cheap: ≤ about one fifth of big banks' CET1 headroom.

A swap: structural buffers cut in 2020 and 2024.

No sign of contraction in credit or its price. Complements ECB tightening; reciprocated abroad; **€6.7bn** releasable.

Verdict: **normalisation, not tightening.**

Trade-off: relies on discretion and credibility; release never tested.

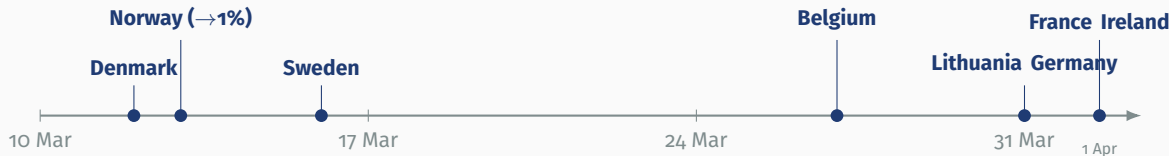
One-pager (Zheng, 0:45) [0:15–1:00]

Here is the whole talk on one slide; you can come back to it at any point.

- In 2022 and 2023, De Nederlandsche Bank, DNB, raised the buffer twice, to two percent, while the Basel rulebook pointed to **zero**.
- Our verdict: this was **normalisation, not tightening**. DNB was refilling a buffer, not fighting a boom.
- It cost banks little, it was largely offset by cuts in other buffers, and we see **no sign of a contraction in lending**.

The rest of the talk is the story of how we got there.

When COVID hit, neighbours released their buffers — the Netherlands had nothing to release



Netherlands: CCyB 0% — nothing to release

DNB improvised (17 Mar 2020):

- cut systemic buffers of ING, Rabobank, ABN AMRO: 3% → 2.5 / 2 / 1.5%
- postponed the mortgage risk-weight floor
- €8bn freed, up to €200bn of lending

... and made a promise

“Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%.”

March 2020 (Zheng, 1:30) [1:00–2:30]

Let's start in March 2020. COVID has just hit Europe.

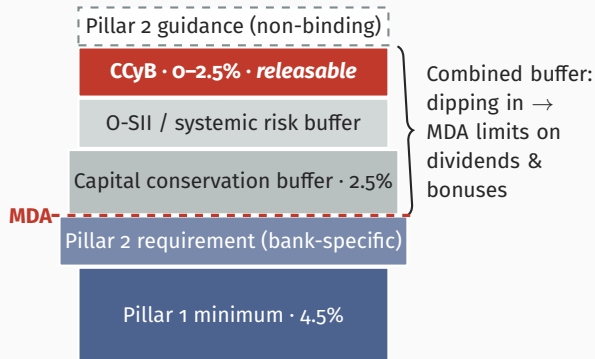
- Within days, Denmark, Norway, Sweden, then Germany, France, Ireland, Lithuania and Belgium **release** their countercyclical buffers.
- **[def]** A capital buffer is extra equity that banks must hold above the legal minimum, so they can absorb losses without failing.
- Releasing it tells banks: use this capital now, keep lending.

The Netherlands could not do that. Its countercyclical buffer was at **zero**, so there was **nothing to release**.

So DNB improvised.

- On 17 March 2020 it cut the systemic buffers of its three largest banks, ING, Rabobank and ABN AMRO, from three percent to 2.5, 2 and 1.5 percent.
- It also postponed a planned floor on mortgage risk weights.
- Together that freed about **eight billion euros** of capital, which DNB said could support up to **200 billion** of lending.

And then DNB made a promise (*pause*): once things were back to normal, it would rebuild this capital as a **two-percent countercyclical buffer**. Keep that promise in mind; the whole story turns on it.



Design

- 0–2.5% of risk-weighted assets, on *domestic* exposures
- Increases bind after 12 months; releases are immediate
- EU reciprocity: foreign banks lending in NL hold it too

Why “releasable” matters

- In COVID, banks close to buffer thresholds **cut lending** even when *allowed* to use buffers (stigma, MDA)
- Only an explicitly **released** buffer is truly usable

The one buffer built to be released (Zheng, 1:00) [2:30–3:30]

Why does it matter *which* buffer you hold? Because not all capital is usable in a crisis.

- **[def]** The countercyclical capital buffer, the CCyB, is the only layer of this stack that the authority can **switch off** overnight.
- It ranges from 0 to 2.5% of risk-weighted assets. **[def]** Those are a bank's assets weighted by their riskiness, so a safe mortgage counts for less than a risky corporate loan.
- Increases take a year to bind; releases are immediate.

The other buffers are technically usable, but if a bank dips into them, it faces automatic limits on dividends and bonuses. **[def]**

That is the "maximum distributable amount", or MDA, restriction. Banks hate that, and markets punish it.

- ECB research shows that in COVID, banks close to these thresholds **cut lending**, even though they were allowed to use the buffers.
- Only a buffer the authority explicitly releases is truly usable.

That is the lesson of COVID. What matters is not only how much capital banks hold, but how much of it the authority can actually hand back.

DNB raised capital requirements in the middle of a storm, when its own rulebook said zero. Wrong moment?

The question

The storm (2022–23)

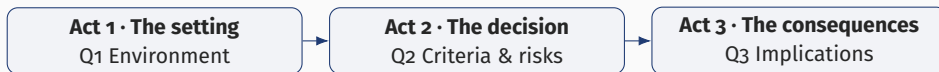
War in Ukraine · record energy prices
HICP inflation **11.6%** (2022)
ECB hikes from **21 Jul 2022**
House prices turning down
Basel gap **-33 / -47pp** → guide **0%**



The decision

CCyB **0% → 1% → 2%**
announced May 2022 and May 2023
binding May 2023 and May 2024

Raising capital in a downturn is the **procyclical mistake** the CCyB was invented to avoid.
Tightening into a slowdown? Fighting a hidden boom? Or something else?



The question (Zheng, 1:00) [3:30-4:30]

Fast-forward to 2022. DNB keeps its promise, but look at the timing.

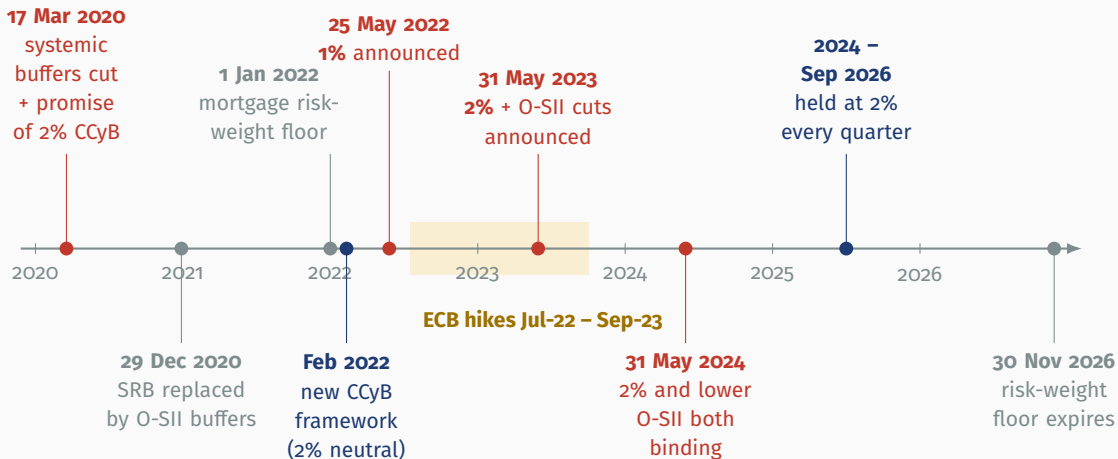
- War in Ukraine, record energy prices, inflation of **11.6 percent**, the ECB raising rates from July 2022, and house prices about to turn down.
- And the Basel indicator says **zero**.

[def] Raising capital requirements in a downturn is *procyclical*: it amplifies the cycle instead of dampening it. That is precisely the mistake the countercyclical buffer was invented to prevent.

So our question is simple: **was this the wrong moment?** We answer it in three acts, which are the three questions of the assignment: the setting, the decision, and its consequences.

The path to 2% was announced in 2020 — and walked step by step inside the ECB's tightening cycle

Q1 • Environment



Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 17 Sep 2026; 24 Apr 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

Timeline (Zheng, 1:00) [4:30–5:30]

First, the setting, beginning with the timeline.

- The path was announced in March 2020.
- In February 2022 DNB published a new framework.
- In May 2022 it announced one percent; in May 2023 two percent, together with lower buffers for systemically important banks.
- Both changes bound on the same day: **31 May 2024**.
- The buffer has been held at two percent every quarter since, most recently in September 2026.

Notice the gold band: the ECB's hiking cycle. Both decisions sit right inside it, which is why the timing question is real.

The economy had fully recovered — then war, energy and inflation darkened the outlook

Q1 • Environment

2021Q3

GDP back above
pre-COVID level

11.6%

HICP inflation, 2022
(national CPI 10.0%)

+50bp

ECB's first hike
21 July 2022

record

energy & commodity prices
after the invasion

DNB, Financial Stability Report, spring 2022

“After a strong recovery from the pandemic, the economic outlook has worsened due to the war in Ukraine and high inflation, and financial stability risks have increased.”

“Higher funding costs may put pressure on the debt sustainability of governments, businesses and households.”

By end-2021 the Dutch economy was almost 3% larger than at end-2019 — the recovery DNB's framework requires before building the buffer.

Macro environment (Zheng, 1:05) [5:30–6:35]

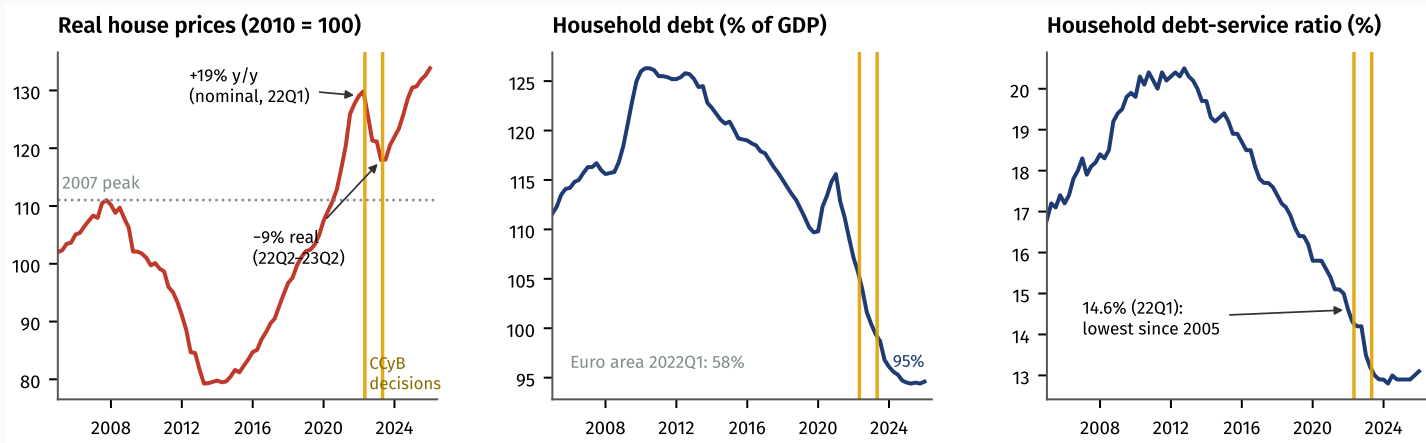
What did the economy look like?

- By the third quarter of 2021, Dutch GDP was back above its pre-COVID level, and by the end of 2021 it was almost three percent larger than before the pandemic. The recovery was complete, and that is the condition DNB's framework requires before building the buffer.
- Then came the storm. In its spring 2022 Financial Stability Report, DNB wrote that "the economic outlook has worsened due to the war in Ukraine and high inflation".
- Energy prices hit records, inflation reached 11.6 percent in 2022, and on 21 July 2022 the ECB raised rates by fifty basis points, its first hike.

So the macro picture gave DNB arguments **both ways**.

- The recovery said: build the buffer now, while the economy can take it.
- The storm said: wait, because higher capital requirements could hurt a weakening economy.

Higher rates also raise the burden on borrowers, and DNB warned about the debt sustainability of governments, firms and households.



Housing: nominal prices +19% y/y (22Q1), real 16% above the 2007 peak. DNB itself: *“the overheated housing market calls for further measures.”*

Credit: household debt high (107% of GDP vs 58% in the euro area) but *falling*; debt service at its lowest since 2005.

Credit and housing (Zheng, 1:15) [6:35–7:50]

In the financial system, the picture was split.

- **Housing was hot.** In early 2022 nominal house prices rose 19 percent year-on-year, and real prices were 16 percent above their 2007 peak. DNB itself called the housing market "overheated".
- But look at what happened next: real prices fell by about nine percent within a year, as mortgage rates rose.
- **Credit was cold.** Household debt is high by international standards, about 107 percent of GDP against 58 percent in the euro area, but it was **falling**.
- **[def]** The debt-service ratio, the share of income that goes on interest and repayments, was at its **lowest since 2005**.

So if there was a boom, it was in house prices, not in credit.

Banks entered the storm well capitalised — and rising rates were lifting their profits

Q1 • Environment

17.7%

CET1 ratio, end-2021
(EU 15.7%, 21Q3)

16.3%

CET1 ratio, end-2022
“in line with EU average”

+6.7%

net interest income 2022
(67.7% of income)

11.5%

four major banks' CET1
after FSR-2023 stress (-3.8pp)

Against raising now

war and energy shock · 11.6% inflation
rapid ECB tightening · house prices turning

For raising now

economy recovered · credit cold
banks strong, profits rising with rates

DNB (FSR 2022): higher rates “*may ultimately have a positive impact on the profitability of financial institutions.*”

Banks (Zheng, 1:05) [7:50–8:55]

Who would carry the losses if things went wrong? The banks, and they were in good shape.

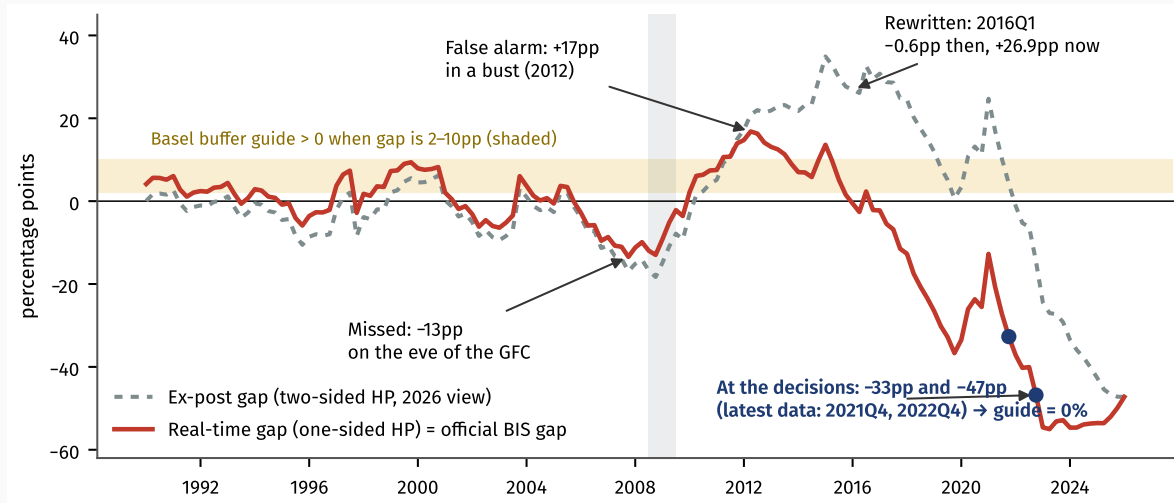
- **[def]** The core capital ratio, CET1, is high-quality equity as a share of risk-weighted assets. It was 17.7 percent at the end of 2021, above the EU average, and 16.3 percent a year later.
- In DNB's stress test, the four major banks would lose 3.8 points of capital but stay well above the minimum.
- Crucially, rising rates were **lifting** bank profits: net interest income grew 6.7 percent in 2022.

DNB noted that higher rates "may ultimately have a positive impact on the profitability of financial institutions". Keep that sentence in mind: it comes back as our twist.

So the setting cuts both ways. Against raising: war, inflation, rate hikes, housing turning. For raising: a recovered economy, cold credit, and strong, increasingly profitable banks.

HAND-OVER: "That was the setting. Diego will now show what DNB actually looked at, and what it was insuring against."

The Basel rule said 0% — but on Dutch data it had missed the last crisis and raised false alarms



Rule: gap < 2pp → 0%, gap > 10pp → 2.5%. Our one-sided HP filter ($\lambda = 400,000$) reproduces the official BIS gap within 0.00005pp (1971–2026).

Alternative trend filters are noisy around COVID (two-sided HP +4.2pp in 2021Q4) — one more reason not to trigger on any of them.

Why not the Basel rule? (Diego, 1:15) [8:55–10:10]

Thank you. Let me start with the rule DNB chose *not* to follow.

- **[def]** The Basel rule looks at the credit-to-GDP gap: how far total credit, as a share of GDP, sits above its long-run trend. That trend is estimated with a statistical smoother called the Hodrick-Prescott filter.
- A gap below two points means a zero buffer; above ten points means the maximum.

We rebuilt this indicator ourselves and match the official BIS series almost exactly. On Dutch data the rule has a poor record:

- **It missed the last crisis:** the gap was minus 13 points on the eve of the financial crisis.
- **It raised a false alarm** in 2012, during a house-price bust, simply because GDP was falling.
- **It gets rewritten after the fact:** in 2016 it read slightly negative in real time, but plus 27 points with today's data.

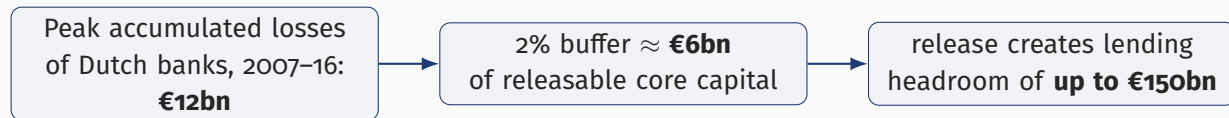
Why does it fail so badly in the Netherlands? Dutch credit rose from under fifty percent of GDP in the 1960s to around 350 percent by 2015. The filter interprets much of that structural deepening as "trend". When the ratio later comes down, as it has since its peak of about 350 percent in 2015, the gap turns deeply negative, even if nothing about cyclical risk has improved.

At the two decisions it read minus 33 and minus 47 points, so the rule said zero. On this record, DNB had good reason to use a different compass.

DNB's compass: 2% is the normal-times level, calibrated on past crisis losses — not on credit growth



built up 1pp a year → 2% after two years



- Criteria: a dashboard (macro, banks, households & firms, markets); *“no mechanical link ... guided discretion.”*
- 2% chosen *“also taking into account the impact of the buffer reduction in March 2020.”*
- Not alone: 17 jurisdictions run a positive neutral CCyB; NL, Sweden, UK, Poland target 2%. ECB loss-based range: 1.1–1.8%.

DNB's framework (Diego, 1:15) [10:10–11:25]

That compass is DNB's 2022 framework, with four phases.

- After a crisis, the buffer is released.
- In **normal** times, it should be at **two percent**, built up one point a year.
- Only when risks are clearly elevated does it go above two percent.
- When a crisis hits, it is released again.

[def] This is a "positive neutral" buffer: a buffer that is already filled when risks are neither high nor low.

How did DNB get to two percent? From history.

- In past crises, Dutch banks' peak accumulated losses were about **12 billion euros**.
- A two-percent buffer is about **six billion** of releasable capital, and releasing it could support up to **150 billion** of lending.
- **[def]** Decisions are taken by "guided discretion": a dashboard of indicators informs, but does not dictate.

The Netherlands is not alone: seventeen jurisdictions now run such a framework, and Sweden, the UK and Poland also target two percent.

DNB justified the steps by recovery, uncertainty and its 2020 promise — not by credit growth

Q2 · Criteria & risks

May 2022 · 0% → 1%

*“When the systemic risk buffers ... were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB.”*

“The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but ... the robust economic recovery ... provides grounds for a gradual build-up.”

Cyclical risks: “normal” to elevated.

May 2023 · 1% → 2%

*“Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk.”*

Cited: investors' rising risk appetite · real-estate prices falling · debt sustainability of firms and governments · banks “robust, partly due to rising interest rates”.

Both decisions: **a normal-to-elevated risk picture + strong banks + the promise.** Credit growth is never the reason.

What DNB said (Diego, 1:00) [11:25–12:25]

What reasons did DNB give?

- **In 2022:** "When the systemic risk buffers were lowered in March 2020, we also announced our intention to restore the buffers by raising the CCyB." DNB pointed to the strong recovery and described risks as "normal to elevated", while acknowledging the uncertainty of the war.
- **In 2023:** DNB said explicitly that the credit gap shows "**no signs of excessive credit growth**". It cited investors' rising risk appetite, falling real-estate prices, debt sustainability, and banks made more robust by higher rates.

So in neither decision was credit growth the reason. The reasons were a normal-to-elevated risk picture, strong banks, and the promise.

The CCyB insures against shocks nobody can forecast; the known hot spots got their own tools

Q2 · Criteria & risks

Financial-stability risks named by DNB (FSR 2022/23)

Instrument

War, energy and supply shocks · rapid tightening of financial conditions and market corrections · debt sustainability of governments, firms, households · bank turmoil abroad (SVB, Credit Suisse, 2023)

CCyB — releasable for *any* shock

“Overheated” housing · high mortgage debt of first-time buyers · interest-only loans

LTV/LTI limits, tax reform, **mortgage risk-weight floor** (Jan 2022 – Nov 2026)

Size and concentration of the large banks (sector **400%** → **280%** of GDP)

O-SII buffer (cut 2024)

Principle: one instrument per target (Tinbergen, 1952)

DNB on the 2020 recomposition: more releasable capital is “*a valuable addition given the sensitivity/volatility of the Dutch economy to external events.*” A releasable buffer is the only tool that works whatever the source of the shock.

Risks addressed (Diego, 1:15) [12:25–13:40]

So which financial-stability risks was the buffer meant to address? DNB's reports list them, and each risk was matched to a tool.

- **Shocks nobody can forecast** go to the countercyclical buffer, because it can be released whatever the source: the war, an energy shock, a sudden tightening of financial conditions, debt-sustainability problems, or bank turmoil abroad like SVB and Credit Suisse in 2023.
- **The known hot spot, housing**, went to other tools: borrower-based limits, [def] caps on loan-to-value and loan-to-income, tax reform, and a floor on mortgage risk weights.
- **The size of the big banks** goes to the buffer for systemically important banks.

[def] This is the Tinbergen principle: one instrument per target. DNB itself said that more releasable capital is valuable "given the sensitivity of the Dutch economy to external events".

If DNB were ...	normalising	fighting overheating	what happened	
Responding to credit indicators	no link	rises with them	gap, debt service, HH debt ratio <i>falling</i> ; CCyB <i>up</i>	✓
Path	+1pp a year, stop at 2%	follows risk	1% → 2% → held	✓
2023: house prices falling, rates jumping	raise anyway	pause	raised	✓
2024–26: house prices rising again (+10% y/y, Mar-26)	stay at 2%	go above 2% (phase 3)	held at 2%	✓
Other buffers	cut structural ones to compensate	keep them	cut in 2020 and 2024	✓

Five predictions, five matches: **normalisation**. The last row is the key to the whole story.

The verdict (Diego, 1:30) [13:40–15:10]

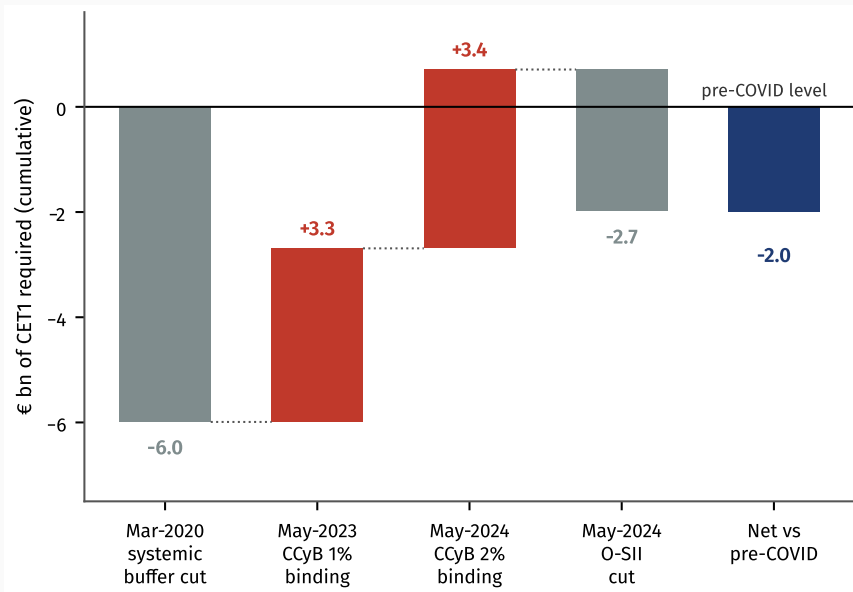
Now we can test our two suspects: was DNB fighting overheating, or normalising? Each makes different predictions. Let's go row by row.

- **Link to credit indicators.** Overheating says the buffer rises with them. In fact they were all falling while the buffer went up.
- **Path.** Normalisation says one point a year, then stop at two. That is exactly what happened.
- **2023.** House prices were falling and rates jumping. An overheating fighter would pause. DNB raised anyway.
- **2024 to 2026.** House prices rose again, about ten percent a year by early 2026. An overheating fighter would go above two percent. DNB held at two.
- **Other buffers.** If you are normalising, you cut the structural buffers to compensate. DNB did, twice.

(pause) Five predictions, five matches.

What is the strongest evidence against us? House prices at plus 19 percent in early 2022. But DNB itself called housing "overheated" and asked for tax and borrowing-rule changes, not a higher countercyclical buffer. And when house prices fell in 2023, it raised the buffer anyway. That is not how you fight a housing boom.

That last row is the key to the whole story.



DNB's own words

"more or less capital-neutral for the three large banks"

— 2020

"a limited increase in the net capital requirements"

— Knot, Jun 2023

"slightly increases the capital requirements for the banking sector as a whole"

— FSR 2023

Not unusual: the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

Waterfall: approximate, own calculation. Buffer cuts valued at end-2022 total RWA of ING, Rabobank, ABN AMRO, de Volksbank (BNG excluded); CCyB amounts are DNB's sector-wide figures on Dutch exposures. Relative to 2021 the change was an increase.

Source: DNB press release 17 Mar 2020; Knot (2023), statement to the Standing Committee for Finance, 7 Jun 2023; DNB FSR spring 2023; BCBS (2024) d585; bank annual reports 2022.

A swap, not a squeeze (Diego, 1:15) [15:10–16:25]

Here is what that last row means in euros. This chart is our own approximate calculation.

- In 2020 the systemic-buffer cut released roughly **six billion** of required capital.
- The countercyclical buffer then added **3.3 billion** in 2023 and **3.4 billion** in 2024.
- At the same time the buffer for systemically important banks was cut by about **2.7 billion**.
- Net, relative to before COVID, the requirement is roughly **unchanged**, if anything slightly lower.

DNB said the same: the 2020 plan would be "more or less capital-neutral". In 2023, Governor Knot spoke of "a limited increase in the net capital requirements".

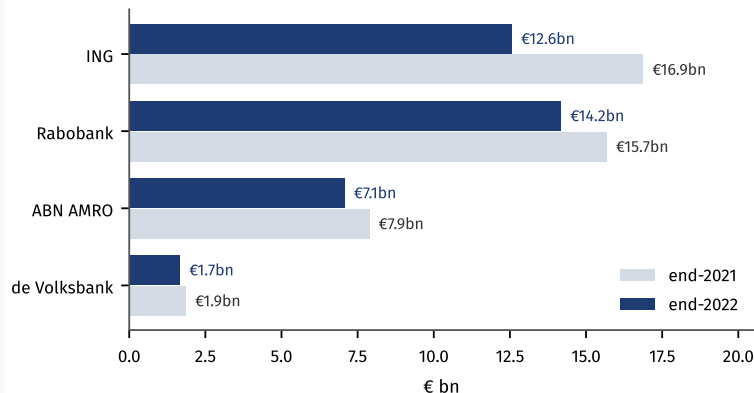
Two caveats on our numbers. The cuts apply to the big banks' worldwide assets, while the countercyclical buffer applies to all banks' Dutch exposures, so the bases differ. And measured from 2021 rather than from before COVID, there was a modest increase. We come back to that.

So capital was **moved**, from structural buffers that cannot be released into one that can. The UK did something similar when it set its two-percent neutral rate.

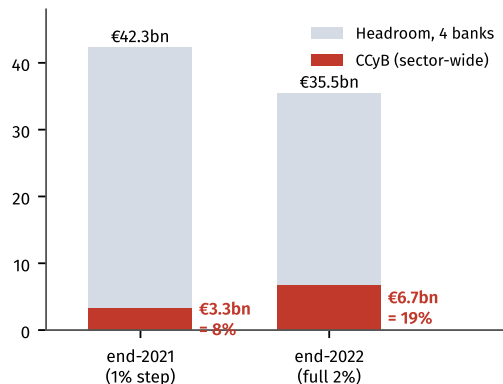
But a swap still has a cost, especially in a storm. So, back to our question: was it the wrong moment?

The twist: the storm made it the cheapest moment — higher rates gave banks the profits to build the buffer

CET1 headroom above requirement / MDA, by bank



CCyB as share of headroom (upper bound)



Net interest income **+6.7%** in 2022 → banks build buffers from *retained earnings*, not by cutting loans. Headroom shrank because of RWA growth and payouts, not the CCyB.

ECB: gradual build-up + high profitability “limit these economic costs” (MPB 24); activating buffers during the 2022–23 tightening “can mitigate implementation costs” (MPB 31).

Not the wrong moment — arguably the right one.

The twist (Diego, 1:15) [16:25–17:40]

Here is the twist: the storm made it the **cheapest** moment.

- **[def]** Look at headroom, the capital banks hold above their requirements. At the end of 2022 the four large banks had about 36 billion euros of it.
- The full two-percent buffer is 6.7 billion, so it absorbs at most about **a fifth** of that headroom, and that is an upper bound.

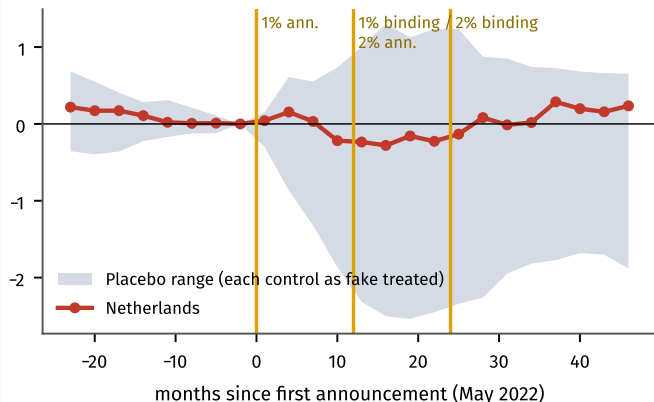
Bank by bank, the left panel shows the same: ING and Rabobank each had well over ten billion euros of headroom, ABN AMRO about seven. Headroom did shrink between 2021 and 2022, but because risk-weighted assets grew and banks paid out capital, not because of the buffer, which did not yet bind.

Why so cheap? Because rising rates lifted net interest income. Profitable banks can build buffers from retained earnings instead of cutting loans.

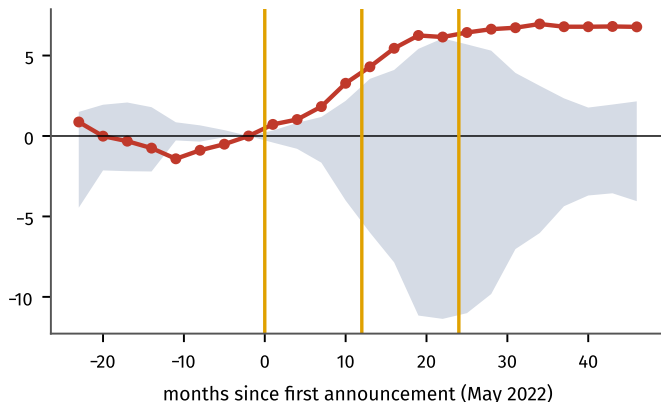
The ECB reached the same conclusion. Its research finds that a gradual build-up with profitable banks "limits these economic costs", and that activating buffers during the 2022–23 tightening "can mitigate implementation costs".

So: not the wrong moment. Arguably the right one.

Mortgage lending rate, NL minus controls (pp)



Household loan growth, NL minus controls (pp)



- Dutch lending rates stayed **inside the placebo range** at all three CCyB dates
- Household credit grew **faster** than in peers; bank credit 2022–26: NL +16.6% vs euro area +7.9% (but NL lagged in 2019–22: +4.3% vs +11.3%)

DNB (Sep 2026): “no signs of constraints in bank credit supply.”

One treated country, small control group (AT, FI, IT, MT, LU), housing rebound: **no sign of contraction**, not a causal estimate.

Borrowers (Diego, 1:15) [17:40–18:55]

Cheap for banks, but did borrowers pay?

- **[def]** We ran an event study: we compare Dutch lending rates and loan growth with euro-area countries that did not change their buffer, around each of the three policy dates.
- **[def]** With only one treated country, we judge significance with a placebo test: we pretend each control country raised its buffer and see how large such "fake" effects are. That is the grey band.

The red line is the Netherlands.

- Its **lending rates stay inside the band**: no unusual change at any of the three dates.
- **Household credit actually grew faster** than in the peer countries.

One more honest detail. Over 2022 to 2026, Dutch bank credit grew 16.6 percent against 7.9 percent in the euro area. But in the three years before, the Netherlands had lagged, 4.3 against 11.3 percent, so part of that later gap is catching up.

Our claim is modest: one treated country and a small control group, so this is "**no sign of contraction**", not a causal estimate. DNB's own verdict in September 2026 was the same: "no signs of constraints in bank credit supply".

HAND-OVER: "So the decision was cheap and credit kept flowing. But a buffer like this reaches beyond Dutch banks and borrowers. Zheng."

Building the buffer while the ECB hiked did not double the squeeze — it let monetary policy focus on inflation

Q3 · Implications

The concern

Two tightenings at once: ECB rates *and* capital requirements → credit crunch?

What the data show

Dutch lending rates did not diverge from euro-area peers (event study); bank profits *rose* with rates (net interest income +6.7%).

One monetary policy for the euro area, but national macroprudential policy: the CCyB is the **Dutch-specific lever** — and the insurance against a rate shock gone wrong.

ECB Macroprudential Bulletin 31 (2025)

Early CCyB activation “helps monetary policy focus on its primary objective of price stability, thereby largely eliminating the potential for conflict between price stability and financial stability.”

DNB, FSR 2022

“A sudden tightening of financial conditions could also have a negative impact on financial stability.”

Monetary and macroprudential policy (Zheng, 1:05) [18:55-20:00]

Thank you. The first wider implication is the interaction with monetary policy.

- The worry was a **double squeeze**: the ECB raising rates while DNB raised capital requirements.
- As we just saw, Dutch lending rates did not diverge from their peers, and bank profits rose with rates.

The ECB argues that building buffers early "helps monetary policy focus on its primary objective of price stability". [\[def\]](#) Macroprudential policy targets the stability of the financial system as a whole, and it does not have to fight inflation.

For a euro-area country this matters even more. There is one monetary policy for twenty countries, but macroprudential policy is national. The countercyclical buffer is the Dutch-specific lever, and it is the insurance in case a rate shock goes wrong.

Through reciprocity every bank lending in the Netherlands holds the 2% — and Europe is moving the same way

Q3 · Implications

Across borders

- Foreign banks with Dutch exposures must also hold the 2% (mandatory reciprocity up to 2.5%)
- Limits leakage to foreign lenders — a known problem with national capital tools (Aiyar, Calomiris & Wieladek, 2014); *non-bank* leakage remains
- In April 2023, 5 banking-union countries had positive-neutral frameworks; NL had the **highest** rate (2%)

CCyB increases in the euro area

Country	Rate	Applies from
Germany	0.75%	Feb 2023
France	1.0%	Jan 2024
Ireland	1.5%	Jun 2024
Netherlands	2.0%	May 2024
Belgium	1.0%	Oct 2024
Portugal	0.75%	Jan 2026
Spain	1.0%	Oct 2026

Across borders (Zheng, 1:00) [20:00–21:00]

Second, borders.

- **[def]** Under EU reciprocity, any foreign bank lending into the Netherlands must also hold the Dutch two percent on those loans.
- That limits leakage, where lending simply moves to foreign banks, a known problem with national capital rules. Lending can still leak to non-banks, though.

And the Netherlands is part of a wider shift: Germany, France, Ireland and Belgium raised their buffers in 2023–24, and Portugal and Spain follow in 2026. In 2023 the Netherlands had the highest neutral rate in the banking union. The ECB's own surveys describe the same shift: authorities now want to build releasable capital early, rather than wait for a credit boom that the indicators may never show.

€0 → €6.7bn

releasable-by-design capital
2020 → 2024

≈ half

of the €12bn peak losses
after the GFC

€150bn

lending headroom
if released

Asymmetric payoff (Lang & Menno, 2023)

Normal times: +1pp requirement → about **-0.1%** lending.

Bad times: 1pp release → up to about **+10%** lending.

Cheap to hold, powerful to release.

A growing role

The mortgage risk-weight floor expires on 30 Nov 2026.
DNB: this “*underpins the importance of ... the current countercyclical capital buffer (CCyB) of 2%.*”

Resilience gained (Zheng, 1:00) [21:00-22:00]

Third, what the Netherlands gained.

- In 2020 it had **no** capital that was releasable by design. Today it has about **6.7 billion euros**, roughly half of the peak losses after the financial crisis, enough to support up to 150 billion of lending.
- ECB research shows why this is a good deal: raising requirements in normal times costs very little lending, about 0.1 percent per point, while releasing in bad times can boost lending by up to ten percent.

And its role is growing. The mortgage risk-weight floor expires at the end of November 2026, and DNB says this "underpins the importance" of the two-percent buffer.

Credibility	Without a mechanical rule, markets must trust DNB's judgement and communication — the classic rules-vs-discretion trade-off (Kydland & Prescott, 1977).
Calibration	Why 2%, not 1.5%? The ECB's loss-based range is 1.1–1.8%; NL chose the top of the banking union.
Untested	No release yet: will banks lend the freed capital — or hoard it, as with usable buffers in COVID?
Baseline	Relative to 2021 (not 2019) it was an increase: +€3.3bn before the O-SII offset.
Same profits, two claims	The ECB lists the Netherlands among countries adopting bank levies on excess profits in 2022–23 — partly claiming the earnings that fund the buffer.

Critiques (Zheng, 1:00) [22:00–23:00]

Fourth, the open questions, because this is not a clean win.

- **Discretion needs credibility.** Without a mechanical rule, markets must trust DNB's judgement. That is the classic rules-versus-discretion trade-off: a rule is predictable but can be wrong, as we saw on slide 9; discretion is flexible but must be explained, every quarter, to stay credible. DNB's quarterly dashboard is its answer.
- **Calibration.** Why two percent? ECB estimates suggest 1.1 to 1.8.
- **It has never been tested:** we do not yet know whether banks will lend released capital or hoard it.
- **Baseline.** Measured against 2021 rather than 2019, it was an increase.
- **Competing claims.** The ECB lists the Netherlands among countries that taxed banks' excess profits in 2022–23, partly the same profits that fund the buffer.

A promise kept: the buffer was refilled in the storm — at the moment it was cheapest

Q1 · Environment

Recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.

Q2 · Criteria & risks

2% normal-times level, calibrated on past losses; insurance against unforeseeable shocks; housing to other tools; structural buffers cut in return.

Q3 · Implications

Cheap · largely a swap · no sign of contraction · complements monetary policy · reciprocated abroad · €6.7bn releasable · relies on discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it — even in a storm.

Back to March 2020 (Zheng, 1:30) [23:00–24:30]

Let me answer the three questions of the assignment.

- **The environment:** a recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.
- **The criteria and risks:** a two-percent normal-times level, calibrated on past losses, insuring against shocks no one can forecast. Housing went to other tools, and structural buffers were cut in return.
- **The implications:** cheap, largely a swap, no sign of contraction, complementary to monetary policy, reciprocated abroad, and 6.7 billion euros now releasable. The price is reliance on discretion.

If there is one idea to take away, it is this: the Dutch decision was less about how *much* capital banks hold, and more about *which kind*. Capital that the authority can release when the next shock comes, whatever its source.

(pause) In March 2020, the Netherlands had nothing to release. **Next time, it will.**

The best time to fill a buffer is when nobody thinks you need it, even in a storm. Thank you; we look forward to your questions.

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

Primary sources

BCBS (2024). Range of practices in implementing a positive neutral CCyB (d585).
 DNB (2020). DNB lowers bank buffer requirements to support lending, 17 Mar.
 DNB (2022). Analytical framework for setting the CCyB in the Netherlands.
 DNB (2022, 2023). CCyB press releases, 27 May 2022, 31 May 2023; O-SII news item, 31 May 2023.
 DNB (2026). CCyB decisions 26 Mar and 17 Sep; risk-weight measure expires, 24 Apr.
 DNB. Financial Stability Report, spring 2022 and spring 2023.
 Knot, K. (2023). Introductory statement, Standing Parliamentary Committee for Finance, 7 Jun.
 ECB (2020). Temporary capital and operational relief, 12 Mar; ECB (2022). Monetary policy decisions, 21 Jul.
 ESRB. CCyB rates table; O-SII notifications NL (27 Nov 2020; 31 May 2023); COVID-19 measures.
 Data: BIS (credit gaps, total credit, DSR, property prices); ECB Data Portal (MIR, BSI); CBS; bank annual and Pillar 3 reports (ING, Rabobank, ABN AMRO, de Volksbank).

Literature

Aiyar, Calomiris & Wieladek (2014). Does macro-prudential regulation leak? *JMCB* 46(s1).
 Behn, Pereira, Pirovano & Testa (2023). A positive neutral rate for the CCyB. *ECB MPB* 21.
 Couaillier, Lo Duca, Reghezza & Rodríguez d'Acuña (2022). Caution: do not cross! *ECB WP* 2644.
 De Nora, Pereira, Pirovano & Stammwitz (2025). From losses to buffer. *ECB WP* 3061.
 Detken, Hempell & Pirovano (2025). Macroprudential and monetary policy interaction. *ECB MPB* 31.
 Drehmann, Borio, Gambacorta, Jiménez & Trucharte (2010). Countercyclical capital buffers: exploring options. *BIS WP* 317.
 Edge & Meisenzahl (2011). The unreliability of credit-to-GDP ratio gaps in real time. *IJCB* 7(4).
 Hamilton (2018). Why you should never use the Hodrick-Prescott filter. *REStat* 100(5).
 Herrera, Scalone & Pirovano (2024). The importance of being positive. *ECB MPB* 24.
 Kydland & Prescott (1977). Rules rather than discretion. *JPE* 85(3).
 Lang & Menno (2023). The state-dependent impact of changes in bank capital requirements. *ECB WP* 2828.
 Repullo & Saurina (2011). The countercyclical capital buffer of Basel III: a critical assessment. CEMFI WP 1102.
 Tinbergen (1952). *On the Theory of Economic Policy*. North-Holland.

Methods and references (no speech; for questions)

Point to this slide if asked about methods:

- HP and Hamilton filters
- headroom formula: CET1 ratio minus requirement, times risk-weighted assets
- event-study design and placebo inference
- all sources listed are verified